CODEHIVE

JAVA PROGRAMMING: OBJECT-ORIENTED FOUNDATIONS

What is OOP?

Object-Oriented Programming (OOP) is a paradigm centered around **Objects** rather than functions. It allows programmers to mirror real-world entities like Cars, Users, or Bank Accounts in their code.

Procedural vs. OOP

Procedural	Object-Oriented (OOP)
Focuses on step-by-step logic	Focuses on objects & data
Functions & Data are separate	Methods & Data are encapsulated
Hard to maintain large apps	Scalable & Modular design

Classes & Objects

A **Class** is the blueprint. An **Object** is the physical instance created from that blueprint.

```java
// The Blueprint (Class)
class Car {
    String brand;
    void drive() { System.out.println("Driving..."); }
}

// The Instance (Object)
Car myCar = new Car();
myCar.brand = "Tesla";
```

The DRY Principle: "Don't Repeat Yourself." OOP enables you to write code once in a class and reuse it across multiple objects, reducing bugs and saving time.

OOP Benefits

- **Reusability:** Create a class once, use it everywhere.
- **Maintenance:** Update one class to fix all related objects.
- **Security:** Protect data through encapsulation.
- **Organization:** Makes large codebases readable.